

Achieving Strategic Objectives Through Operational Modeling

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Abstract - An organization achieves competitive advantage by right strategic positioning and operational effectiveness. For the organization, the cost of its products and services is influenced by the effectiveness of the business processes that drive the organization. Effectively managing those processes, in the context of key performance measures, can reduce the cost and thereby ensuring organizational effectiveness. In this paper, we introduce a framework to identify the most effective actions that help translate strategy to organizational effectiveness.

I. INTRODUCTION

Whether a company sells a product or service, all of its business processes exist for one primary reason, which is to drive the sale. In other words, the cost of any sale is the cumulative effect of all the business processes in the organization. For instance, if the cost of hiring an employee is high, then the cost of the sale goes up.

Assuming that the price of the product or service cannot be increased due to factors such as market conditions and the competitive landscape, the only variables under management's control to increase the profit is to reduce the cost of sales and/or increase the volume of sales. How can good profits be ensured? The answer lies in effective Business Process Management (BPM) implemented through the right choice of technology. However, amongst the myriad of business processes in an organization, it is critical to select the right ones to manage to drive global organizational effectiveness.

Note that reducing the cost of making a product or service has to be done in the context of other performance measures. For instance, investing in research and development may increase the cost in the short-term, but will potentially have long-term benefits such as creating innovative products, thereby assuring future revenue.

As mentioned above, there are two factors that increase profits on the sale of a product or a service: (a) reduce the cost of the sale; and, (b) increase the volume of the sale. On one end of the spectrum, the likes of Dell and Wal-Mart, that have small margins and move large volumes of inventory, have achieved operational effectiveness through supply chain optimization and logistics. On the other end of the spectrum, the likes of McKinsey provide a high-margin service for a relatively smaller number of clients, but their operational effectiveness is achieved through content and knowledge shared across projects. In either case, it can be argued that the operational effectiveness of the organization's business processes contributes toward higher profits.

We develop a methodology to identify the critical business processes that have to be tuned for effectiveness, to achieve the best return on investment. Such candidate processes may fall out of an exercise that transforms strategic initiatives to actions. For instance, if the company strategy is to increase customer satisfaction, then all processes that directly touch the customer are candidates. These may include the order fulfillment process, customer services process, and product returns process. We further narrow this list by selecting those that are most critical and currently the worst performing.

The Balanced Scorecard [1] offers a mechanism to identify and track various performance indicators to gauge the health of an organization. In particular, the methodology outlines how to identify the operational performance metrics in the four categories of finance, customer satisfaction, internal business processes, and learning and growth. However, it does not explicitly lay down the steps to identify, for example, the specific internal business processes that are critical to the organizational effectiveness. Nor does the methodology discuss a systematic way to identify and prioritize those actions that will be most effective in helping the organization achieve its performance objectives.

We propose a framework for identifying and prioritizing those actions required to meet the business performance goals. In this approach, first we determine the current state of the organization. Then we determine the goal of the organization, based on its strategic plan. After that, we identify the actions required to achieve the goals that are in line with the strategic objectives. Finally, we offer a mechanism to prioritize those actions. Given the limitation on the resources, it is often not possible to execute all the actions required to increase performances. The usefulness of this framework is to cull down the list of candidate business processes to optimize, manage, and measure, resulting in high-return actions.

II. CROSS ORGANIZATIONAL DEPENDENCIES

Most business processes, especially the candidate ones that need to be managed, will span multiple business units. For instance, the order fulfillment process spans sales, finance, inventory management, and shipping, at the very least. The person who leads the charge on tuning such cross-functional process improvements, therefore, needs to have the support of, and visibility into, these functional units and should be able to enforce cross-functional cooperation. To effectively do that, this person needs to understand the

fundamentals of BPM and have the domain knowledge to manage its effective implementation.

Additionally, BPM initiatives require synergy between business and information technology (IT) units, since most process improvements are implemented through technology with the business facilitating the collaboration. Such a synergy can only be achieved when the business managers understand what IT can or cannot accomplish, and ensure that IT is aligned with the business.

Any business process improvement requires the support and sponsorship of senior executives and managers. However, many managers do not understand their own processes because they operate at a high-level. At the other end, the operational people may understand only their particular component in the process. For instance, the shipping clerk may have no idea of how a product is ordered or how it gets into the inventory. This results in nobody understanding the “architecture” of the process. A process architect position may be created, if it does not already exist, to understand the overall process.

One way for the stakeholders to understand their processes would be to participate in a simulated process activity such as an order [2]. A “simulation coordinator” will ensure that process is simulated effectively, and the result is that each participant understands the overall process, its efficiencies, and inefficiencies, to drive the project to success.

III. IDENTIFY CANDIDATES

There are many ways to identify candidate business processes to tune. We discuss two of them here. Whichever is used, the objective is to identify a small set of business processes that can move the company from the “as-is” state to the “to-be” state.

A. Based on Strategy

One way to identify candidates is to use business objectives that we want to achieve. The “as-is” measure gives us an indication of where the organization is, and the “to-be” is a measure of where the organization ought to be. The “to-be” measure is often derived from the strategy. For instance, we may want to improve two measures, order fulfillment turnaround time and quality, to achieve competitive advantage. Figure 1 shows the “as-is” and “to-be” states using these indicators and a suitable metric.

The “X” in Figure 1 shows where the organization stands in terms of these two performance measures, and “O” shows where it would like to be. This organization wants to move from a 15-day turnaround time and below-average quality, to a 5-day turnaround time and above-average quality.

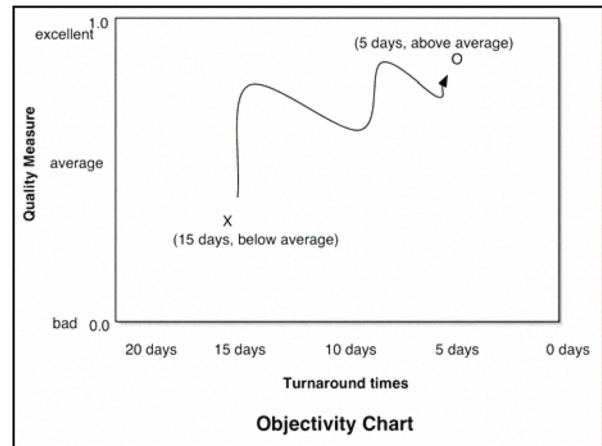


Figure 1: Objectivity Chart

Since the number of business objectives that we want to optimize happens to be two (turnaround time and quality), the objectivity chart is two-dimensional. In reality, businesses use ten to fifteen measures, some of which may be derived from the Balanced Scorecard.

Management has to decide what needs to be done operationally, to move from point X to point O. Perhaps adding more automation to the production process will yield better quality. Perhaps switching to a different shipping company for delivery will reduce the turnaround time.

B. Based on Cost

Another way to identify candidate business processes for tuning is to identify those that contribute to the cost of the product or service that the company is selling.

On a sale, the maximum profit occurs when the cost price is the least. A high profit margin alone is not sufficient. The sales volume is also a factor in determining the dollar amount of profit. Working backwards, this means that if we want \$X in profits this year, anticipate selling Y units, and the selling price is fixed, then we can calculate the maximum cost price we can incur per unit. Continuing along those lines, we can identify the business processes that have to be tuned to achieve that objective.

A note of caution is required here. Every business process that exists in the organization contributes to the cost of the sale. While this does mean that every process is a candidate for tuning, evaluating each becomes a daunting task. In addition, the list of all business processes in the organization is rarely documented. It may be useful to heuristically cull this list to a more manageable number.

While it will be ideal to know all the factors that contribute towards the cost of a product or service, this data is often elusive to obtain. Given that's the case, a good starting point would be to get the approximate contributions of the high-level activities in the organization, and then work on reducing those that may yield the biggest "bang-for-the-buck." The lack of the data has to be made up by inputs from the stakeholders, which can be contentious at times.

IV. QUANTIFY EVERYTHING

In an organization, there are two kinds of measures: quantitative and qualitative. Our goal is to make all measures quantitative to reduce ambiguity. The number of defects found on a product can be quantified, and when done so, there is often little argument about its validity, if done properly. However, qualitative measures are subjective. For instance, "company goodwill" is a qualitative measure. If one of the strategic objectives is to increase the company goodwill, then it has to be measurable to determine progress. Unfortunately, since no real number can be associated with it, it has to be done subjectively. The range of subjectivity can be constrained narrowly by taking a collaborative discussion-oriented approach so that each participant understands the perspectives of others. The Delphi Method [3] is one such formalization of a structured approach to collaborative forecasting and estimation. We discuss an adaptation of this method in Section VII.

Let's say we want to measure n parameters that represent the business performance of a company. Borrowing from the principles of pattern recognition [4], if each of these measures can be quantified, then they can be represented as a point C in multidimensional space, which we refer to as the *performance space*, as follows:

$$C = (x_1, x_2, \dots, x_n)$$

where x_i is the numeric value corresponding to the i^{th} performance metric. As we discussed earlier, not all metrics are quantifiable, and those that are not will have to be quantified using the Delphi Method.

Now assume that through a series of collaborative discussions, perhaps using the Balanced Scorecard as the framework, the management team identifies a set of actions to be taken to accomplish the strategic objectives of the organization. Such actions should help move the organization to another point, F , in the performance space, thus:

$$F = (y_1, y_2, y_3, \dots, y_n)$$

Points C and F are the current and future business performance indicators respectively. The organization

has to move from point C to point F , and the actions to accomplish such a movement have also been determined. Executing these actions should take the organization closer to achieving its strategic objectives. The *Strategic Objective* is the "distance" between the current state and the desired state, and is given by:

$$\text{Strategic Objective: } SO = \sqrt{\sum (x_i - y_i)^2}, \text{ for } i=1, n$$

If the *performance space* is normalized in the range [0,1] as it should be, then the SO is an indicator of how much work is left to achieve the business objective.

V. ACTIONS FOR EXECUTION

One principle of strategic management [5] is achieving competitive advantage by either performing different processes than competitors, or performing similar processes in different ways.

Let's say that the actions required to achieve the strategic objectives are $A_1, A_2, A_3, \dots, A_m$. However, some A_i that increases a certain performance metric may decrease another performance metric. For instance, putting more money into R&D may result in innovative products, but will at the same time increase the cost of operations. This in fact may be a conflicting goal in the short term, but has potential long-term benefits by providing for products in the future. Any action is pursued to achieve a certain objective. Our objectives have already been stated as F , which is $(y_1, y_2, y_3, \dots, y_n)$. The actions required to achieve the strategic objectives are not independent of one another. The next step is to determine the dependencies between these actions.

Just like the qualitative and quantitative metrics, such dependencies can be determined by data analysis, or where that is lacking, through the Delphi Method. The result is a dependency matrix that shows the degree of dependency amongst the various actions.

The dependency matrix will look like this:

	A_1	A_2	...	A_m
A_1		0.2	0.3	
A_2				
...				
A_m	-0.4			

From this matrix, we know that A_2 has a dependency on A_1 with strength of 0.2. The strength indicates the degree by which one action affects another. On the other hand, there is a negative correlation between A_m and A_1 with a strength of 0.4. This indicates that when action A_m is undertaken, then the effect of action A_1 will decrease.

We can represent the effect of the actions on other actions and the objectives using the model shown in Figure 2.

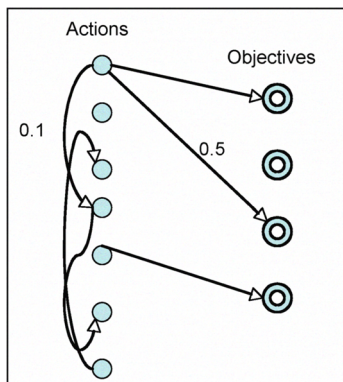


Figure 2: Effect of actions on business objectives

The model shows both the dependencies of the actions with other actions, as well as its contribution to the business objectives. Plugging into the model, the fact whether an action is taken or not shows us the result of these actions in terms of the strategic objectives. Using this model, for example, we could determine the ROI on implementing a particular action. For instance, the cost of one action may be too high relative to another one for the same ROI. In such a case, we may want to choose the one with the least cost to prioritize first. On the other hand, one parameter of the actions could be driven by customer needs, in which case it may take a higher priority.

Note that in some cases, the numeric value of y_i may actually be less than the numeric value of x_i . This may simply indicate that the metric is negatively correlated with performance, such as the cost of manufacturing a widget (the lower the cost, the better), or it may indicate that the organization does indeed want to lower the metric. For instance, as a percentage of revenue, the research budget needs to be reduced.

VI. THE TIME DIMENSION

There is a time lag between when a particular action is taken and the objectives achieved. While each action may contribute to multiple objectives, each may occur at a different point in time. Let's say that a business decides to take money from its R&D budget and apply it to a marketing campaign. Right after the start of the campaign, the consumers' interest in the company may be high, but may eventually wear out. However, since money was taken away from R&D, the long-term effects of that result may include a smaller product set in the future.

This means that the effect of a certain action, A_p on an objective, O_q , has to be modeled along the time

dimension as well. The action of a marketing campaign on the sales volume, if that is one of the objectives, may be modeled over the time dimension as shown by the curve in Figure 3.

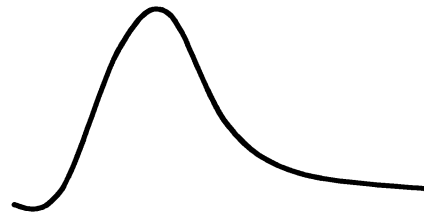


Figure 3: Marketing campaign's effect over time

In other cases, the effect of a particular action may follow a simple step function. For instance, if a firm spends \$X on hiring ten people, there will still be ten people at the firm for a long time (assuming 100% retention). Similarly, other behavioral patterns are possible.

This means that for each action, not only do we associate a weight that represents its contribution to a certain objective, but we also need to add its effect over time. Doing so will result in the following time-effect matrix;

	O_1	O_2	...	O_k
A_1		$f(A_1, O_2)$	$f(A_1, O_3)$	
A_2				
...				
A_m	$f(A_m, O_1)$			

Where the A s represents the actions, the O s represent the objectives and $f(A_i, O_j)$ represents the variation of objective O_j over time because of a certain action, A_i .

Using the model, we can then determine the effects of the actions on the objectives over a period of time. Such a model is useful as a management tool to make decisions on what actions to undertake to meet the business objectives in the most effective manner.

VII. USING THE DELPHI METHOD

We use the Delphi Method to accomplish a number of things:

- To get the stakeholders to communicate with one another and use it as a tool for each to understand the other persons' perspective.
- To use it as a framework for discussions to arrive at a consensus and to document all the associated points of discussion. The result, therefore, tends to be based on a logical conclusion.
- To convert a qualitative metric to a quantitative metric, which is then useful for

modeling the effectiveness of actions on the business processes and eventually the performance indicators of the organization.

- d) To agree upon the performance indicators to use.
- e) To identify the list of actions required to achieve the strategic objectives of the organization.

In the Delphi Method, various stakeholders are brought together to arrive at a consensus. For example, in the first round to quantify the performance of a certain performance measure, the participants will first offer their own perspectives on how the company is ranked on that measure. As an example, one person may grade the quality of a product at 0.4 while another might grade it at 0.9, on a normalized [0,1] scale. Many people will offer their own grades, each of which falls in the [0,1] range. The variance of these numbers may be large.

In subsequent discussions, the goal is to bring this variance down to a pre-determined number, at which point the goal of quantifying a subjective measure has been achieved. Therefore in each discussion, the participants explain why they graded the way they did, and this brings out and perspectives of different people. As the groups' understanding of the issues converges with subsequent discussions, the grades they assign will also converge into a narrower range.

The exercise, as mentioned above, will also help to understand the issue at hand at a deeper level. An experienced person should facilitate such an exercise to guide the discussions in the right direction. This person will also set some ground rules to ensure a democratic process and serve as the referee.

Just as the Delphi Method is used to quantify subjective measures, it can also be used to identify the critical performance indicators that represent the current "as-is" condition of the organization. It is a suitable tool to leverage for setting the "to-be" state of the organization.

Finally, the Delphi Method can be used to identify the list of actions required to achieve the strategic objectives of the organization. As mentioned earlier, there are multiple ways to determine the actions, such as those based on cost or those based on the organization's strategy, or it may be driven by a need to address customer satisfaction, which may be a high-priority issue. If the business strategy does not incorporate a customer satisfaction component when indeed the customer satisfaction is poor, then it is indicative of a strategy formulation problem, which we will not discuss in this paper.

In a normal Delphi Method, all the participants contribute to all the data to reach a consensus.

However, here there is a possibility that some people will not be interested in some processes. For instance, the VP of Manufacturing may be indifferent to the hiring process or may not have much knowledge about it to comment either way. This effectively means that the attainment of each metric or business objective may be debated only by a smaller subset of people in the room. Thus, conclusions based on such prioritization must be normalized to ensure that all business processes are treated equally.

Note that whatever the Delphi Method is used for, the results need not be linearly weighted. This means that if, through the exercise the team comes up with a list of business processes that have to be tuned, not all of them may be of equal importance. Therefore, a concept of a weight can be applied to the result. These weights can also be determined as part of the Delphi Method. For instance, most people will agree that the customer-centric processes are important because those contribute to the perception of the organization by the customer.

VIII. CONCLUSION

Profits increase by reducing the cost of the sale and increasing the volume of sales. Either of these is driven by the internal business processes in the organization. When an organization sets a strategic goal to achieve business objectives, certain actions are required to achieve those objectives. In this paper, we propose a methodology by which these actions can be identified and prioritized to increase profits.

Using this methodology, we introduce steps to determine and quantify business objectives, and a process to identify the actions to be taken by the organization to achieve those objectives. We did this by creating a model for analysis of the effects of an action on the effectiveness the business processes.

We introduced a modified version of the Delphi Method to derive quantitative measures from qualitative ones, to aid in measuring progress towards predetermined performance objectives. We showed how the method could be used by the team for a common understanding of the issues and for effective execution of the strategy.

We introduced an operational model for a business that shows how an interrelated set of decision variables interact to achieve business objectives to create sustainable competitive advantage.

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